

Ansys Fluent Simulation Report

Analyst	Hp
Date	8/21/2026 01:33 PM

Table of Contents

- [1 System Information](#)
- [2 Geometry and Mesh](#)
 - [2.1 Mesh Size](#)
 - [2.2 Mesh Quality](#)
 - [2.3 Orthogonal Quality](#)
- [3 Simulation Setup](#)
 - [3.1 Physics](#)
 - [3.1.1 Models](#)
 - [3.1.2 Material Properties](#)
 - [3.1.3 Cell Zone Conditions](#)
 - [3.1.4 Boundary Conditions](#)
 - [3.1.5 Reference Values](#)
 - [3.2 Solver Settings](#)
- [4 Run Information](#)
- [5 Solution Status](#)
- [6 Plots](#)
- [7 Contours](#)
- [8 Vectors](#)

System Information

Application	Fluent
Settings	2d, pressure-based, standard k-epsilon
Version	24.1.0-10184
Source Revision	5b3f9fb3c8
Build Time	Nov 22 2023 10:29:19 EST
CPU	11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz
OS	Windows

Geometry and Mesh

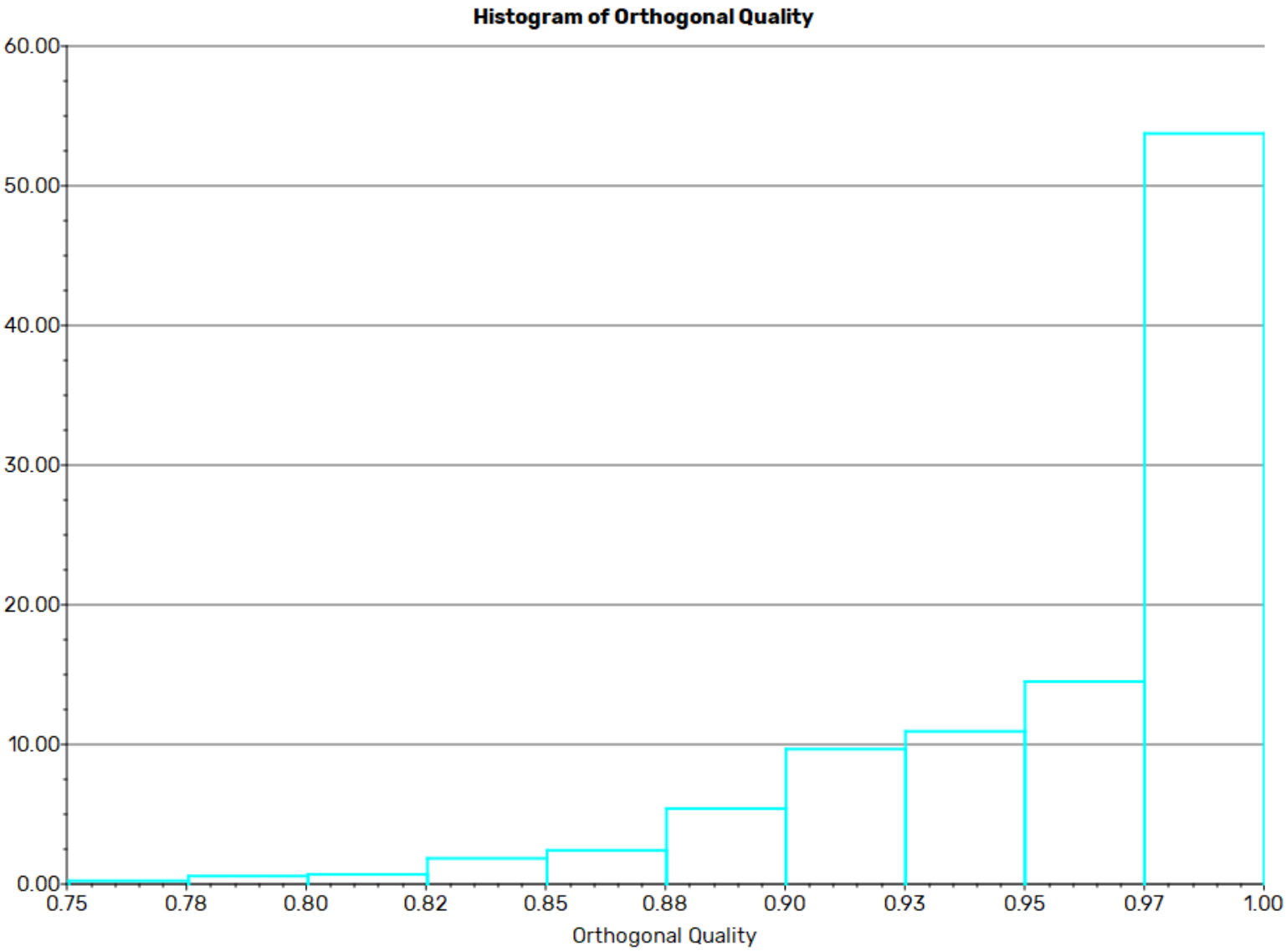
Mesh Size

Cells	Faces	Nodes
278	591	313

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
fff_surface	Mixed Cell	0.7504738	2.9368265

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	2D
Time	Steady
Viscous	Standard k-epsilon turbulence model
Wall Treatment	Standard Wall Functions

Material Properties

— Fluid	
— air	
Density	1.225 kg/m^3
Cp (Specific Heat)	1006.43 J/(kg K)
Thermal Conductivity	0.0242 W/(m K)
Viscosity	1.7894e-05 kg/(m s)
Molecular Weight	28.966 kg/kmol
— Solid	
— aluminum	
Density	2719 kg/m^3
Cp (Specific Heat)	871 J/(kg K)
Thermal Conductivity	202.4 W/(m K)

Cell Zone Conditions

— Fluid	
— fff_surface	
Material Name	air
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Laminar zone?	no
Porous zone?	no

Boundary Conditions

— Inlet	
— inlet	
Velocity Specification Method	Magnitude, Normal to Boundary
Reference Frame	Absolute

Velocity Magnitude [m/s]	10
Supersonic/Initial Gauge Pressure [Pa]	0
Turbulent Specification Method	Intensity and Viscosity Ratio
Turbulent Intensity [%]	5
Turbulent Viscosity Ratio	10
— Outlet	
— outlet	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	0
Pressure Profile Multiplier	1
Backflow Direction Specification Method	Normal to Boundary
Turbulent Specification Method	Intensity and Viscosity Ratio
Backflow Turbulent Intensity [%]	5
Backflow Turbulent Viscosity Ratio	10
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
— Wall	
— cylinder_wall	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
— bottom	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5
— top	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Roughness Height [m]	0
Wall Roughness Constant	0.5

Reference Values

Area	1 m^2
Density	1.225 kg/m^3
Depth	1 m
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	1 m/s

Viscosity	1.7894e-05 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	fff_surface

Solver Settings

— Equations	
Flow	True
Turbulence	True
— Numerics	
Absolute Velocity Formulation	True
— Pseudo Time Explicit Relaxation Factors	
Density	1
Body Forces	1
Turbulent Kinetic Energy	0.75
Turbulent Dissipation Rate	0.75
Turbulent Viscosity	1
Explicit Momentum	0.5
Explicit Pressure	0.5
— Pressure-Velocity Coupling	
Type	Coupled
Pseudo Time Method (Global Time Step)	True
— Discretization Scheme	
Pressure	Second Order
Momentum	Second Order Upwind
Turbulent Kinetic Energy	First Order Upwind
Turbulent Dissipation Rate	First Order Upwind
— Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000
Minimum Turb. Kinetic Energy [m^2/s^2]	1e-14
Minimum Turb. Dissipation Rate [m^2/s^3]	1e-20
Maximum Turb. Viscosity Ratio	100000

Run Information

Number of Machines	1
Number of Cores	1
Case Read	12.236 seconds
Iteration	0.128 seconds
AMG	0.0419996 seconds
Virtual Current Memory	0.411934 GB
Virtual Peak Memory	0.733086 GB
Memory Per M Cell	303.927

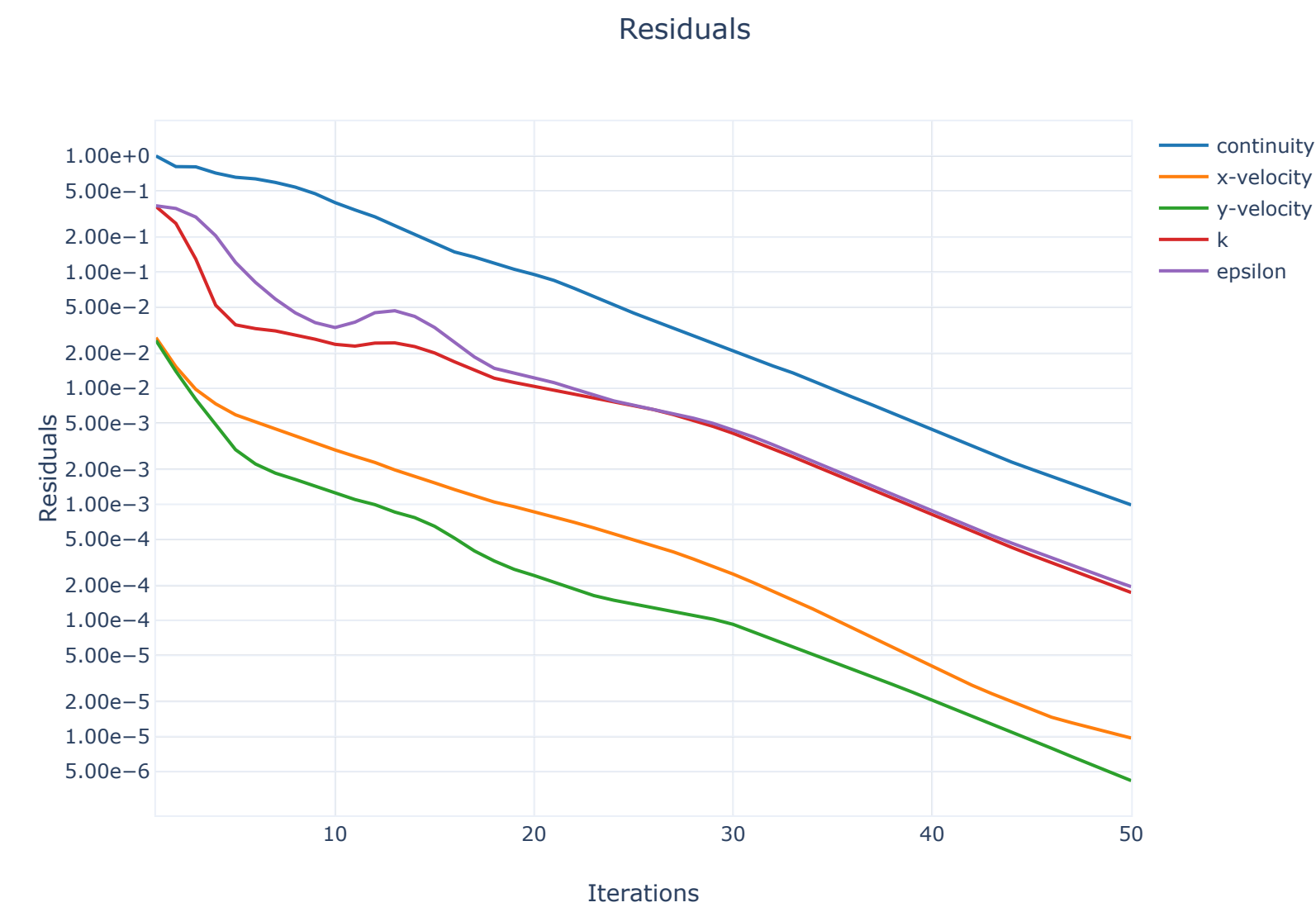
Solution Status

Iterations: 50

	Value	Absolute Criteria	Convergence Status
continuity	0.0009895276	0.001	Converged
x-velocity	9.716555e-06	0.001	Converged
y-velocity	4.172584e-06	0.001	Converged
k	0.0001740172	0.001	Converged
epsilon	0.0001954016	0.001	Converged

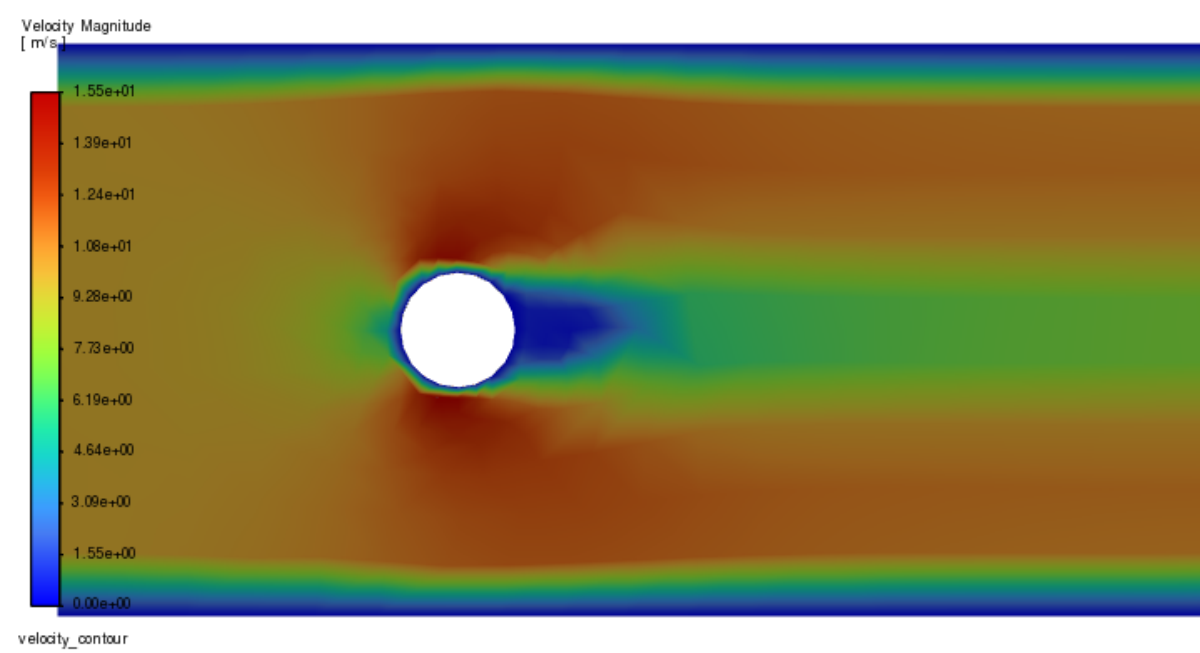
Plots

Residuals

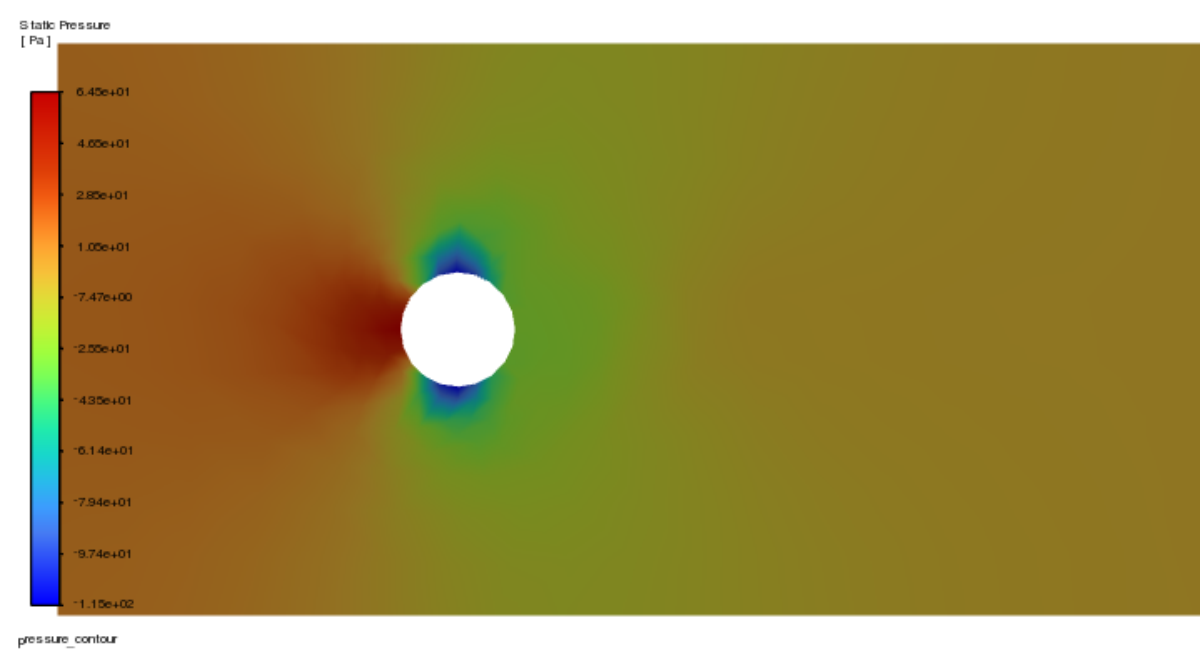


Contours

velocity_contour



pressure_contour



Vectors

pressure_vector

